Digitization and the Middle Income Trap

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1 Introduction

Middle-income countries face a persistent challenge of sustaining growth after reaching moderate levels of development—a phenomenon widely known as the Middle-Income Trap (MIT). While early-stage growth can be driven by capital accumulation, labor reallocation, and export-led industrialization, sustaining long-term convergence toward high-income status requires continuous gains in productivity, innovation, institutional quality, and financial efficiency. In recent years, financial technology (FinTech) and broader digital transformation have emerged as potentially transformative forces capable of reshaping financial intermediation, expanding access to capital, improving resource allocation, and accelerating technological diffusion. Yet, the role of FinTech and digitization in helping economies escape—or inadvertently deepen—the Middle-Income Trap remains insufficiently understood.

This paper investigates the extent to which FinTech adoption and digital financial development influence countries' ability to overcome structural stagnation and transition to high-income growth trajectories. We argue that FinTech can act as a catalyst for productivity-enhancing investment, innovation-driven entrepreneurship, and financial inclusion, thereby strengthening the channels through which economies upgrade technologically and structurally. At the same time, we recognize potential risks: uneven digital adoption, regulatory gaps, platform concentration, and financial fragility may amplify inequality or misallocate capital, limiting the developmental benefits of digitization.

Using cross-country panel data and novel indices of FinTech penetration and digital financial depth, this study examines how digital financial infrastructure interacts with productivity growth, credit allocation, firm dynamics, and institutional quality in middle-income economies. We explore whether FinTech facilitates efficient capital reallocation toward high-value sectors, enhances small- and medium-enterprise (SME) financing, reduces transaction costs, and supports innovation-led growth. The analysis further evaluates heterogeneity across regions and institutional environments to identify conditions under which FinTech most effectively contributes to escaping the Middle-Income Trap.

This paper contributes to three strands of literature: (i) the economics of the Middle-Income Trap and structural transformation, (ii) financial development and growth, and (iii) the emerging literature on FinTech and digital finance in developing economies. By integrating digital financial transformation into the MIT framework, the study offers new empirical evidence and policy-relevant insights on how middle-income countries can harness FinTech to accelerate productivity, deepen financial markets, and sustain long-term economic upgrading.

2 Literature

The relationship between financial development and economic growth has long been central to development economics. Early empirical contributions document a robust positive association between financial depth and long-run growth (King and Levine, 1993; Levine, 2005). Financial systems promote growth by mobilizing savings, allocating capital efficiently, facilitating risk diversification, and supporting technological innovation. However, subsequent work identifies nonlinearities in this relationship. Arcand, Berkes, and Panizza (2015) argue that beyond a threshold, financial deepening may exert diminishing or even negative effects on growth, giving rise to the "too much finance" hypothesis. These findings suggest that the quality, structure, and regulatory context of financial development are critical determinants

of its developmental impact.

Parallel to this literature, the Middle-Income Trap (MIT) framework examines why many economies experience persistent growth slowdowns after reaching moderate income levels. Eichengreen, Park, and Shin (2013) document systematic growth slowdowns at middle-income thresholds, emphasizing productivity deceleration and structural rigidities. Felipe, Abdon, and Kumar (2012) further characterize prolonged middle-income episodes as associated with limited industrial upgrading and export diversification. Doner and Schneider (2016) highlight institutional fragmentation and weak state capacity as central constraints preventing sustained structural transformation. Despite the importance of finance in supporting innovation and industrial upgrading, the MIT literature has largely underexplored the role of digital financial transformation.

The rapid expansion of financial technology (FinTech) introduces new mechanisms that may alter traditional finance–growth dynamics. Micro-level evidence on digital payments and mobile money demonstrates significant welfare gains through reduced transaction costs and improved risk sharing. Jack and Suri (2014) show that mobile money in Kenya enhances households' ability to smooth consumption in response to income shocks. Suri and Jack (2016) document long-run poverty reduction and occupational shifts linked to mobile money access, particularly among female-headed households. These findings establish the micro-foundations of digital finance: improved liquidity, reduced remittance costs, and enhanced resilience to shocks. However, the macroeconomic implications of these mechanisms remain less clear.

Beyond payments, FinTech has reshaped credit markets through algorithmic underwriting, alternative data usage, and platform-based lending. Fuster et al. (2019) show that FinTech mortgage lenders in the United States reduce processing time and expand refinancing activity. Claessens, Frost, Turner, and Zhu (2018) emphasize that FinTech and BigTech entry into financial services may increase competition and inclusion while also raising concerns about market concentration and systemic risk. The Bank for International Settlements

(BIS, 2018) documents rapid growth in FinTech credit globally, noting its potential to reduce information asymmetries but also its vulnerability to regulatory arbitrage.

From a productivity perspective, digital finance may influence capital allocation efficiency. The misallocation literature demonstrates that distortions in capital and labor allocation significantly reduce total factor productivity in emerging economies (Hsieh and Klenow, 2009). If algorithmic lending and digital financial platforms improve screening and reduce credit constraints for high-productivity firms, FinTech could enhance allocative efficiency and support structural upgrading. Yet cross-country empirical evidence directly linking FinTech penetration to sectoral productivity growth remains limited.

At the macro-financial level, the stability implications of FinTech are actively debated. Philippon (2016) argues that despite technological progress, the cost of financial intermediation has not declined proportionately, raising questions about efficiency gains. More recent work emphasizes that digital finance may introduce new systemic risks through interconnected platforms, cybersecurity vulnerabilities, and rapid credit expansion outside traditional banking systems (FSB, 2017; IMF, 2022). Analogous to the broader finance–growth nonlinearities identified by Arcand et al. (2015), digital finance may exhibit threshold effects in which moderate adoption enhances resilience while excessive or weakly regulated growth amplifies fragility.

Institutional quality appears to condition the developmental impact of financial innovation. Acemoglu, Johnson, and Robinson (2005) argue that inclusive institutions underpin long-run growth by providing secure property rights and predictable governance. In middle-income economies characterized by institutional fragmentation, the effectiveness of digital finance may depend on regulatory capacity and bureaucratic professionalism. Digital payment systems can strengthen state capacity through improved tax collection and transparent public transfers, yet weak governance may allow risk accumulation or rent extraction within digital platforms. This interaction between technology and institutions aligns with broader findings in institutional economics emphasizing conditional development outcomes.

Despite the rapid expansion of research on digital finance, few studies integrate FinTech into the structural transformation and Middle-Income Trap frameworks. Existing literature provides strong micro-level evidence on welfare gains (Jack and Suri, 2014; Suri and Jack, 2016), growing analysis of FinTech credit and market structure (Claessens et al., 2018; BIS, 2018), and extensive documentation of middle-income growth slowdowns (Eichengreen et al., 2013; Felipe et al., 2012). However, systematic cross-country evidence examining whether digital financial penetration influences the probability, duration, or severity of middle-income stagnation remains scarce. In particular, the potential nonlinear interaction between FinTech adoption, institutional quality, and productivity upgrading has not been rigorously evaluated.

Clarifying these linkages is crucial for understanding whether digital finance represents a new engine of convergence or merely reshapes existing financial structures without altering long-run growth trajectories. By situating FinTech within a structural development framework, the literature can move beyond access-based measures of inclusion and instead evaluate its role in productivity allocation, innovation financing, and institutional coordination—core determinants of sustained escape from middle-income stagnation.

3 Empirical Framework

Let i index countries and t index years. The objective is to examine whether digital financial development (FinTech) affects (i) growth performance in middle-income ranges and (ii) the likelihood and duration of middle-income stagnation. Let $FinTech_{it}$ denote a measure of digital financial penetration, and let X_{it} denote a vector of standard macroeconomic controls including human capital, trade openness, inflation, investment, population growth, and financial depth. Country fixed effects μ_i and time fixed effects λ_t are included to control for unobserved heterogeneity and global shocks.

The baseline growth specification is given by:

$$g_{it} = \alpha + \beta FinTech_{it-1} + \Gamma' X_{it-1} + \mu_i + \lambda_t + \varepsilon_{it}, \tag{1}$$

where g_{it} denotes real GDP per capita growth. Regressors are lagged to mitigate simultaneity concerns and reflect adjustment dynamics.

To directly connect to the Middle-Income Trap framework, we estimate the probability of a growth slowdown episode. Let $Slowdown_{it}$ be an indicator equal to one if country i experiences a statistically significant decline in trend growth while in the middle-income range. The discrete-choice specification is:

$$Pr(Slowdown_{it} = 1) = \Phi\left(\eta + \theta FinTech_{it-1} + \Pi' X_{it-1} + \mu_i + \lambda_t\right), \tag{2}$$

where $\Phi(\cdot)$ denotes the cumulative normal distribution.

To assess whether FinTech accelerates exit from middle-income status, we estimate a discrete-time hazard model. Let $Exit_{it}$ equal one if country i transitions to high-income status in year t, conditional on being middle income previously:

$$Pr(Exit_{it} = 1 \mid Exit_{i,t-1} = 0) = \Phi\left(\kappa + \psi FinTech_{it-1} + \Omega' X_{it-1} + \tau_t\right), \tag{3}$$

where τ_t captures duration dependence.

Because theory suggests complementarities between digital finance and institutional capacity, we estimate interaction effects:

$$g_{it} = \alpha + \beta FinTech_{it-1} + \delta Inst_{it-1} + \xi(FinTech_{it-1} \times Inst_{it-1}) + \Gamma' X_{it-1} + \mu_i + \lambda_t + \varepsilon_{it}. \tag{4}$$

A positive ξ indicates that the growth effect of FinTech is amplified in stronger institutional environments.

To explore mechanisms, we implement mediator regressions. For each proposed channel

k, let M_{it}^k denote a mediator variable. The first-stage equation is:

$$M_{it}^k = a_k + \pi_k FinTech_{it-1} + \Lambda_k' X_{it-1} + \mu_i + \lambda_t + u_{it}^k. \tag{5}$$

The second-stage equation includes both FinTech and the mediator:

$$Y_{it} = b_k + \beta_k FinTech_{it-1} + \vartheta_k M_{it-1}^k + \Psi_k' X_{it-1} + \mu_i + \lambda_t + e_{it}^k, \tag{6}$$

where Y_{it} denotes growth, slowdown probability, or exit probability. Evidence for a mechanism is consistent when π_k and ϑ_k are statistically significant and when the magnitude of β_k attenuates relative to the baseline.

We focus on three principal channels.

First, the allocative efficiency channel. Mediators include total factor productivity growth and sectoral productivity indicators. FinTech is hypothesized to reduce capital misallocation and improve productivity upgrading.

Second, the structural transformation channel. Following McMillan and Rodrik (2011), aggregate productivity growth is decomposed as:

$$\Delta P_{it} = \sum_s \theta_{is,t-1} \Delta p_{ist} + \sum_s p_{is,t-1} \Delta \theta_{ist}, \tag{7}$$

where p_{ist} is sectoral productivity and θ_{ist} is the sectoral employment share. The second term captures the reallocation (between-sector) component and serves as a mediator of structural change.

Third, the resilience channel examines whether FinTech reduces macroeconomic volatility and dampens the impact of external shocks. We estimate:

$$g_{it} = \alpha + \beta FinTech_{it-1} + \chi Shock_{it} + \omega(FinTech_{it-1} \times Shock_{it}) + \Gamma' X_{it-1} + \mu_i + \lambda_t + \varepsilon_{it}. \tag{8}$$

A positive ω implies that FinTech mitigates the adverse growth effects of shocks.

Finally, to address persistence and potential endogeneity, we estimate a dynamic specification:

$$g_{it} = \rho g_{i,t-1} + \beta FinTech_{it-1} + \Gamma' X_{it-1} + \mu_i + \lambda_t + \varepsilon_{it}. \tag{9}$$

This equation can be estimated using appropriate dynamic panel techniques to account for lagged dependence and simultaneity bias.

Together, these specifications allow us to evaluate the direct effect of digital financial development on middle-income dynamics and to identify the allocative efficiency, structural transformation, and resilience channels through which FinTech may influence long-run convergence.

3.1 FinTech Construction

To quantify digital financial development across countries, we construct a composite FinTech penetration index using country-level indicators from the Global Findex database. The objective is to capture the multidimensional nature of digital financial transformation, encompassing both access to and usage of digital financial services.

Conceptual Framework. Consistent with the empirical framework outlined in Section 2, digital finance is expected to affect growth and middle-income dynamics through three primary channels: (i) improved allocative efficiency, (ii) enhanced structural transformation, and (iii) greater macroeconomic resilience. To reflect these mechanisms, the index incorporates measures of both digital financial inclusion (access) and transaction digitization (usage).

Specifically, we employ four indicators:

1. **Digital Payment Usage (DP)**: Share of adults who made or received a digital payment in the past year.

2. **Mobile Money Account Ownership (MA)**: Share of adults with a mobile money account.

3. **Digital Merchant Payments (MP)**: Share of adults who used digital means to pay merchants.

4. **Digital Account Engagement (DA)**: Share of adults accessing financial accounts through digital channels.

These variables collectively capture the breadth and depth of digital financial intermediation. Payment usage reflects transaction-level digitization and reduced intermediation frictions; mobile accounts capture inclusion and entry into formal digital finance; merchant payments reflect retail and SME formalization; and digital account engagement measures active participation in digital financial ecosystems.

Standardization and Aggregation. Because the four components are measured in percentage terms but exhibit different variances and cross-country dispersion, we standardize each variable prior to aggregation. For each component k, we compute:

$$Z_{k,it} = \frac{X_{k,it} - \bar{X}_k}{\sigma_k}, \tag{10}$$

where $X_{k,it}$ denotes the raw indicator for country i in year t, and $\bar{X}_k$ and σ_k denote the global sample mean and standard deviation, respectively.

The composite FinTech index is constructed as the equally weighted average:

$$FinTech_{it} = \frac{1}{4}\left(Z_{DP,it} + Z_{MA,it} + Z_{MP,it} + Z_{DA,it}\right). \tag{11}$$

Equal weighting is adopted for transparency and to avoid imposing arbitrary importance on any single dimension. This approach ensures that the index reflects broad digital financial penetration rather than being driven by a single component.

The resulting index is standardized with mean zero and unit variance. A one-unit increase in $FinTech_{it}$ therefore corresponds to a one-standard-deviation increase in overall digital financial penetration. In the growth regressions, the coefficient on $FinTech_{it-1}$ measures the effect of a standard deviation increase in digital financial development on subsequent economic performance. To ensure that results are not sensitive to the weighting scheme, we also construct an alternative index using principal component analysis (PCA). The first principal component explains the majority of the common variance across the four indicators and produces qualitatively similar results, confirming that the composite index captures a coherent underlying dimension of digital financial development.

For mechanism analysis, we further decompose the index into access and usage components:

$$Access_{it} = \frac{1}{2}\left(Z_{MA,it} + Z_{DA,it}\right), \tag{12}$$

$$Usage_{it} = \frac{1}{2}\left(Z_{DP,it} + Z_{MP,it}\right). \tag{13}$$

This decomposition allows us to distinguish whether middle-income escape is primarily driven by expanded access to digital finance or by deeper transactional usage within the economy.

Figure 1 illustrates the evolution of Singapore's FinTech index. Digital financial penetration rises steadily between 2014 and 2017, followed by a pronounced acceleration after 2017 and a sharp increase through 2024. The convex trajectory suggests nonlinear diffusion and strong ecosystem complementarities in advanced institutional environments.

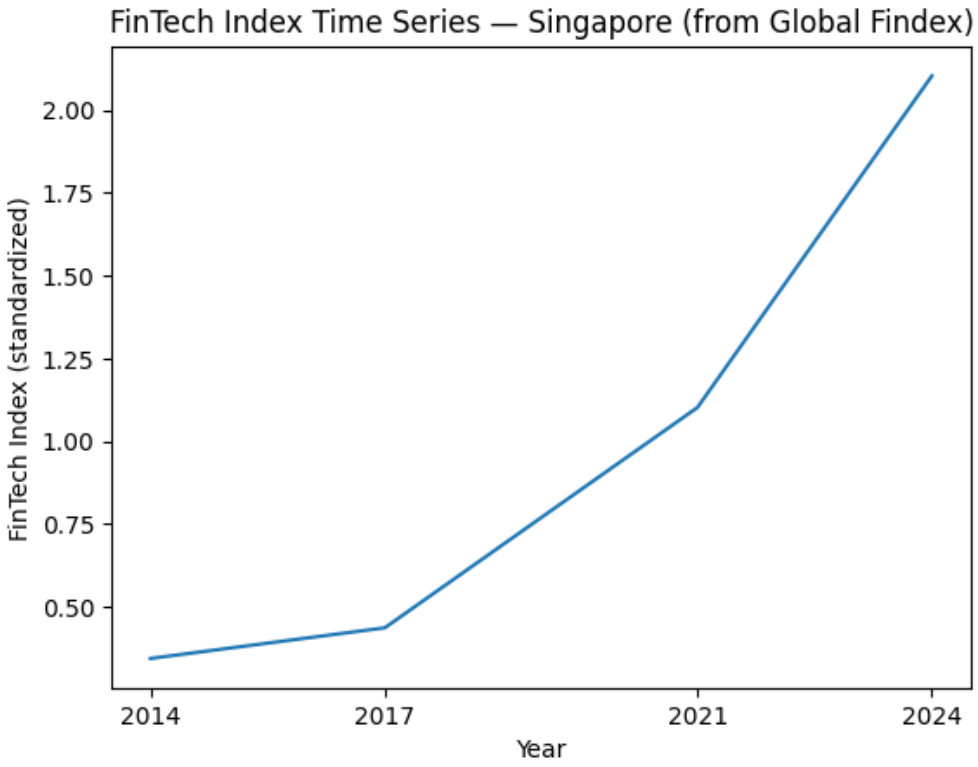


Figure 1: FinTech Score for Singapore

3.2 A Simple Model of FinTech, Credit Frictions, and the Middle-Income Trap

This section develops a stylized framework linking FinTech adoption to productivity growth and middle-income escape through the relaxation of financial frictions. The model features households, firms, and an intermediary sector. The central mechanism is that digital finance reduces borrowing frictions and improves the allocation of capital toward more productive firms. These gains are amplified by stronger institutional quality.

Households. A representative household supplies labor inelastically and saves through the financial system. Lifetime utility is given by

$$U = \sum_{t=0}^{\infty} \beta^t u(C_t), \tag{14}$$

where $0 < \beta < 1$ is the discount factor and C_t denotes consumption. Household savings are deposited with financial intermediaries and transformed into loans to firms. Let total savings be denoted by S_t.

Firms. The economy consists of a continuum of firms indexed by $j \in [0, 1]$. Each firm produces according to

$$Y_{jt} = A_{jt} K_{jt}^{\alpha} L_{jt}^{1-\alpha}, \tag{15}$$

where A_{jt} denotes firm-specific productivity, K_{jt} is capital, and L_{jt} is labor. Firms differ in productivity, and high-productivity firms generate higher returns to capital. In the absence of frictions, capital would flow toward firms with higher A_{jt}.

Each firm has internal funds N_{jt} and must borrow to finance investment:

$$K_{jt} = N_{jt} + B_{jt}, \tag{16}$$

where B_{jt} denotes external borrowing.

Credit Frictions. Because of asymmetric information, weak documentation, poor credit records, and limited screening capacity, firms face borrowing constraints. Let the effective borrowing limit be

$$B_{jt} \leq \phi(F_t, I_t) N_{jt}, \tag{17}$$

where $\phi(\cdot)$ is an endogenous leverage factor that depends positively on FinTech penetration F_t and institutional quality I_t:

$$\phi_F > 0, \qquad \phi_I > 0, \qquad \phi_{FI} > 0. \tag{18}$$

FinTech raises ϕ by reducing information costs, digitizing payment records, expanding

transaction histories, and improving screening technologies. Institutional quality raises ϕ by improving contract enforcement, data reliability, regulatory oversight, and creditor protection. The complementarity term $\phi_{FI} > 0$ captures the idea that digital finance is more effective in stronger institutional environments.

Intermediation Costs. Let firms borrow at interest rate

$$r_{jt} = r_t^* + \tau(F_t, I_t), \tag{19}$$

where r_t^* is the benchmark funding cost and $\tau(F_t, I_t)$ is the intermediation wedge. We assume

$$\tau_F < 0, \qquad \tau_I < 0, \qquad \tau_{FI} < 0. \tag{20}$$

Thus, greater FinTech penetration and stronger institutions reduce the effective cost of external finance. This can occur through lower transaction costs, faster payments, reduced verification expenses, and improved monitoring.

Firm Optimization. Each firm chooses K_{jt} and L_{jt} to maximize profits:

$$\max_{K_{jt}, L_{jt}} \Pi_{jt} = A_{jt} K_{jt}^{\alpha} L_{jt}^{1-\alpha} - w_t L_{jt} - r_{jt} B_{jt}, \tag{21}$$

subject to the borrowing constraint above.

In a frictionless benchmark, capital would be allocated so that the marginal product of capital is equalized across firms. Under credit frictions, however, productive firms may remain underfunded, while less productive firms survive due to distortions. This generates capital misallocation and lowers aggregate productivity.

Aggregate Productivity and Allocation. Aggregate output is

$$Y_t = \int_0^1 Y_{jt}\, dj. \tag{22}$$

Aggregate total factor productivity depends not only on the distribution of firm-level productivity A_{jt}, but also on the efficiency of capital allocation across firms. Let aggregate productivity evolve according to

$$A_{t+1} = A_t \left(1 + \eta_1 F_t + \eta_2 F_t I_t - \eta_3 \mathcal{M}_t\right), \tag{23}$$

where $\mathcal{M}_t$ denotes the degree of capital misallocation. FinTech reduces $\mathcal{M}_t$ by relaxing borrowing constraints for high-productivity firms and expanding access to formal financial services. Stronger institutions amplify this effect by making digital information credible and enforceable.

Middle-Income Trap Mechanism. A middle-income trap arises when productivity growth becomes too weak to sustain convergence after low-wage growth drivers are exhausted. In the model, stagnation occurs when financial frictions remain sufficiently severe that productive firms cannot scale, innovation is underfunded, and capital remains trapped in low-productivity uses.

Let the probability of escaping the middle-income trap be increasing in sustained productivity growth:

$$Pr(Escape_t = 1) = \Psi(A_{t+1} - A_t), \tag{24}$$

with $\Psi'(\cdot) > 0$. Since FinTech raises productivity growth by reducing borrowing costs and improving allocative efficiency, it increases the probability of escape. This effect is stronger when institutional quality is high.

Table 9.3 reports the results of the regression analysis.

Table 1: FinTech Development and the Probability of Remaining in the Middle-Income Trap

	(I)	(II)	(III)
FinTech Index (std.)	0.482***	0.214	0.301*
	(0.145)	(0.162)	(0.180)
Government Effectiveness (std.)	−0.528***	−0.291*	−0.102
	(0.150)	(0.170)	(0.185)
FinTech × Governance		−0.447***	−0.318**
		(0.125)	(0.140)
GDP per capita (log, std.)			−2.284***
			(0.550)
Secondary enrollment (std.)			0.142
			(0.310)
Trade openness (std.)			0.051
			(0.110)
Manufacturing share (std.)			0.421**
			(0.185)
Constant	−0.981***	−0.702***	−1.887***
	(0.110)	(0.150)	(0.330)
Observations	165	165	120
Pseudo R^2	0.061	0.129	0.244